Session 01: Significant Figures & Scientific Notation

Topic: Learning to speak the language of measurement precision.
Time: 120 minutes
Big Fat Notebook: Chapters 1–2

English	Pronunciation & Meaning
significant figures (sig figs)	/sɪɡˈnɪf.ɪ.kənt ˈfɪɡ.jərz/ — digits that carry meaning about measurement precision
precision	/prɪˈsɪʒ.ən/ — how closely repeated measurements agree with each other
accuracy	/ˈæk.jə.rə.si/ — how close a measurement is to the true value
scientific notation	/ˌsaɪ.ənˈtɪf.ɪk noʊˈteɪ.ʃən/ — writing numbers as $a \times 10^n$ where $1 \leq a < 10$
decimal places	/ˈdes.ə.məl pleɪ.sɪz/ — digits after the decimal point; used in +/– rounding
rounding	/ˈraʊn.dɪŋ/ — reducing digits to match the least precise measurement
leading zero	/ˈliː.dɪŋ ˈzɪr.oʊ/ — zeros before the first nonzero digit; never significant
trailing zero	/ˈtreɪ.lɪŋ ˈzɪr.oʊ/ — zeros after the last nonzero digit; significant only with a decimal point

---

Opening Attack

Try this:

> A student measures a cube's edge as **3.40 cm**. Another student measures the same cube's edge as **3.4 cm**. Both students calculate the volume (edge^3). The first student gets **39.3 cm^3**. The second student gets **39 cm^3**.
>
> Both used the same cube. Both did the math correctly. Why are their answers different — and which one is right?

Stuck? Good. Let's figure out why.

Worked Pattern

Example 1 — Counting Significant Figures

How many significant figures are in each measurement?

Measurement	Count	Why?
3.40 cm	**3**	Nonzero digits (3, 4) are always significant. The trailing zero after the decimal counts — it was measured.
3.4 cm	**2**	Nonzero digits only. No trailing zero was recorded, so it wasn't measured.
0.00450 g	**3**	Leading zeros (0.00) are placeholders — they don't count. The 4, 5, and trailing zero after decimal all count.
2500 mL	**2** (ambiguous)	Without a decimal point, the trailing zeros might or might not be measured. In Honors Chemistry, treat this as 2 sig figs unless told otherwise.
2500. mL	**4**	The decimal point says: "All four digits were measured."
6.022×10^{23}	**4**	Only the coefficient (6.022) determines sig figs. The exponent is exact.

Example 2 — Multiplying and Dividing: Count Sig Figs

Problem: A rectangle measures 12.3 cm by 5.6 cm. Calculate its area.

Step 1: Multiply the numbers.

> $12.3 \times 5.6 = 68.88$

Step 2: Count sig figs in each measurement.

> $12.3\ \text{cm} \to$ 3 sig figs
>
> $5.6\ \text{cm} \to$ 2 sig figs $\leftarrow$ **limiting**

Step 3: The answer gets the smaller count: 2 sig figs.

> $68.88 \to 69\ \text{cm}^2$ (2 sig figs)

Answer: $\boxed{69\ \text{cm}^2}$ (limited by 5.6 cm, which has only 2 sig figs).

Example 3 — Multiplying and Dividing: Another One

Problem: Density = mass ÷ volume. Mass = 24.50 g. Volume = 8.0 mL. Calculate density.

Step 1: Divide.

> $24.50 \div 8.0 = 3.0625$

Step 2: Count sig figs.

> $24.50\ \text{g} \to$ 4 sig figs
>
> $8.0\ \text{mL} \to$ 2 sig figs $\leftarrow$ **limiting**

Step 3: Round to 2 sig figs.

> $3.0625 \to 3.1\ \text{g/mL}$

Answer: $\boxed{3.1\ \text{g/mL}}$ (2 sig figs).

Example 4 — Adding and Subtracting: Decimal Places Rule

Problem: Add 12.34 g + 0.5 g + 3.210 g.

Step 1: Line up the decimal points and add.

$$12.34 + 0.5 + 3.210 = 16.050$$

Step 2: Look at decimal places (digits after the decimal point).

> 12.34 → 2 decimal places
>
> 0.5 → 1 decimal place ← **limiting**
>
> 3.210 → 3 decimal places

Step 3: Round the answer to 1 decimal place.

$$16.050 \rightarrow 16.1 \text{ g}$$

Answer: $\boxed{16.1 \text{ g}}$ (1 decimal place, limited by 0.5).

> **Key distinction:** Adding/subtracting uses **decimal places**, NOT sig figs. Multiplying/dividing uses **sig figs**. These are two different rules — don't mix them up.

Example 5 — Adding and Subtracting: Another One

Problem: Subtract 25.0 mL – 3.25 mL.

Step 1: Subtract.

$$25.0 - 3.25 = 21.75$$

Step 2: Count decimal places.

> 25.0 → 1 decimal place ← **limiting**
>
> 3.25 → 2 decimal places

Step 3: Round to 1 decimal place.

> $21.75 \to 21.8 \text{ mL}$

Answer: $\boxed{21.8 \text{ mL}}$ (1 decimal place).

Example 6 — Scientific Notation: Writing

Problem: Write 0.000340 in scientific notation with correct sig figs.

Step 1: Move the decimal point until exactly one nonzero digit is to the left.

> $0.000340 \ \to$ move decimal 4 places right $\to \ 3.40$

Step 2: Write as: coefficient × 10^{exponent}.

> Moving right → negative exponent.
>
> 3.40×10^{-4}

Step 3: Check: the coefficient (3.40) shows the sig figs: **3**.

Answer: $\boxed{3.40 \times 10^{-4}}$ (3 sig figs).

Example 7 — Scientific Notation on Your Calculator

Entering 6.022×10^{23}:

> Do NOT type: `6.022 × 10 ^ 23` (slow and error-prone).
>
> Use the **EE** or **EXP** key: `6.022 EE 23`
>
> The display shows: `6.022E23` or `6.022 × 10²³`
>
> This means 6.022×10^{23} — the "E" stands for "×10^".

Practice: Enter 1.67×10^{-27} → press `1.67 EE (-) 27` → displays `1.67E-27`.

⚠️ THE TRAP — Rounding Too Early

The wrong way:

> A block has mass 15.8 g and volume 6.0 mL. Calculate density.
>
> *Wrong work:* 15.8 ÷ 6.0 = 2.63333... → "I'll round now to 2.6" → Then later: 2.6 × (some other number)... $= \text{wrong final answer}$

Why is this wrong? Rounding before the final answer discards information. The calculator holds 10+ digits — use ALL of them until the very end.

The right way:

> **Step 1:** 15.8 ÷ 6.0 = 2.63333... (keep all digits on the calculator)
>
> **Step 2:** × 3.00 (next step) = 7.90
>
> **Step 3:** Now round to 2 sig figs (from 6.0) → **7.9 g**

Spot the difference: Intermediate values stay unrounded. Only the final answer gets rounded.

What We Just Did — The Sig Fig Procedure

Step 1: Count the sig figs (or decimal places) in each measured number.

> **×/÷:** count **sig figs**.
>
> **+/−:** count **decimal places**.

Step 2: Do the math on your calculator. Keep ALL digits — do not round yet.

> The calculator has 10+ digits of precision. Use all of them.

Step 3: Round the final answer to match the LEAST precise measurement.

> **×/÷:** smallest number of **sig figs**.
>
> **+/−:** smallest number of **decimal places**.

Copy this. You will use it for every problem in the drills.

💡 Pro Tips — Calculator Shortcuts & Mental Checks

Tip 1 — The Pacific-Atlantic rule. If a decimal point is **Present**, start counting sig figs from the **Pacific** (left) side — skip leading zeros, then count everything. If the decimal is **Absent**, start from the **Atlantic** (right) side — skip trailing zeros, then count everything.

> Pacific (decimal Present): 0.00450 → skip 0.00, count 450 → 3 sig figs.
>
> Atlantic (decimal Absent): 2500 → skip 00 from right, count 25 → 2 sig figs.

Tip 2 — ×/÷ sig fig shortcut. Find the measurement with the fewest sig figs. That's it. Your answer gets that many. Don't overthink it. 12.3×5.6 → 5.6 has 2 sig figs → answer has 2 sig figs. Done in 2 seconds.

Tip 3 — +/− decimal place shortcut. Find the measurement with the fewest digits after the decimal. That's how many decimal places your answer gets. $12.34 + 0.5$ → 0.5 has 1 decimal place → answer gets 1 decimal place.

Tip 4 — Scientific notation always has exactly one digit before the decimal. $3450 = 3.450 \times 10^3$, not 34.50×10^2. The coefficient must satisfy $1 \leq a < 10$. This is the fastest way to check if you did it right.

Tip 5 — EE key doubles your exam speed. On a timed test, typing `6.022 EE 23` instead of `6.022 × 10 ^ 23` saves 3–5 seconds per calculation. Over 20 problems, that's a full minute saved. Use the EE key every time.

Basic Drills (B1–B10)

> Goal: Automate sig fig counting and rounding. Finish all 10 smoothly.

B1. How many sig figs? (a) 0.0570 (b) 400 (c) 400.0 (d) 9.10×10^3

B2. Multiply: 6.42×3.1. Give the answer with correct sig figs.

B3. Divide: $45.00 \div 6.0$. Give the answer with correct sig figs.

B4. Add: $2.3 + 0.56 + 1.449$. Give the answer with correct decimal places.

B5. Subtract: $15.0 - 3.27$. Give the answer with correct decimal places.

B6. Write in scientific notation with correct sig figs: (a) 0.0000890 (b) 23,400 (3 sig figs)

B7. A student measures mass as 5.60 g and volume as 2.0 mL. Calculate density with correct sig figs.

B8. A rectangle is 14.55 cm by 2.0 cm. Calculate the perimeter (add all four sides) and area. Which rule applies to each?

B9. (Two-part) Part 1: A block has mass 125.0 g and volume 50.0 mL. Calculate density. Part 2: Multiply that density by 2.0 mL to find mass of a smaller piece. Use correct sig figs at each final answer.

B10. (Speed round) Count sig figs in under 1 minute: (a) 0.03040 (b) 1.200 (c) 5000 (d) 5.000×10^3 (e) 0.001 (f) 2.50×10^{-2}

Advanced Drills (A1–A10)

> Goal: Exam-level. Time pressure. Traps. Multi-step integration.

A1. A cylinder has radius 1.25 cm (measured) and height 10.0 cm (measured). Volume of a cylinder = $\pi r^2 h$. π is exact (infinite sig figs). Calculate volume with correct sig figs.

A2. A lab group measures three masses of the same sample: 4.52 g, 4.55 g, 4.48 g. The true mass is 4.50 g. (a) Calculate the average. (b) Are the measurements precise? (c) Are they accurate? (d) If a fourth measurement was 5.10 g, how would precision and accuracy change?

A3. (Trap-laden) A student calculates: $(12.3 + 0.45) \times 2.0$. They add first: $12.3 + 0.45 = 12.75$, then multiply: $12.75 \times 2.0 = 25.5$. Then round to 1 decimal place (from the addition rule) → 25.5. Is this correct? If not, where did they go wrong?

A4. (Reverse engineering) A student divides 15.0 by 3.00 and gets 5.000. Their partner says it should be 5.00. Who is right? Show the correct calculation and explain the mistake.

A5. (Constructive) Write an exam problem where the answer has exactly 3 sig figs and requires a multiplication followed by an addition. Solve it.

A6. (Constructive) Write a problem where a student would get the wrong answer by using the ×/÷ rule when they should use the +/– rule. Show both the wrong and right solutions.

A7. (Data table) Three students each measure the mass of a 10.00 g standard weight five times:

	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5
Student A	9.98	10.01	9.99	10.00	10.02
Student B	10.45	10.44	10.46	10.45	10.45
Student C	9.8	10.3	10.1	9.6	10.4

Which student is precise but not accurate? Which is accurate but not precise? Which is both? Which is neither?

A8. (Timed — under 3 min) The speed of light is 2.998×10^8 m/s. Light takes 8.3 minutes to reach Earth from the Sun. Calculate the Earth-Sun distance in kilometers with correct sig figs. (Watch the units: minutes → seconds, meters → km.)

A9. (Hardest variation) A student measures the radius of a sphere as 2.0 cm. They calculate volume = $\frac{4}{3}\pi r^3$. The $\frac{4}{3}$ and π are exact. (a) What is the volume with correct sig figs? (b) If they had measured the radius as 2.00 cm, how would the volume change? (c) Why does a small change in sig figs cause a bigger effect in volume?

A10. (Exam boss) A student determines density five times: 2.71, 2.65, 2.68, 2.73, 2.60 g/cm^3. The accepted value for aluminum is 2.70 g/cm^3. (a) Calculate average density with correct sig figs for each operation. (b) Calculate percent error: $|\text{avg} - \text{accepted}|/\text{accepted} \times 100\%$. (c) Are the measurements more precise or more accurate? (d) The student realizes the balance was tared with a 0.05 g error. Does this affect precision, accuracy, or both?

Now Read

You can now count sig figs, round correctly after ×/÷ and +/−, write scientific notation, and distinguish precision from accuracy — at exam speed.

Read **Chapters 1–2** of *Everything You Need to Ace Chemistry in One Big Fat Notebook*.

The concepts — why measurements have uncertainty, how sig figs communicate that uncertainty, why precision ≠ accuracy — will lock into place. You've solved 20 problems harder than anything the book asks.

> Solutions: solutions/01-solutions.md